Chapter 16

Optimization Problems and Antiderivatives

§ Section: 4.7,4.9

16.1 Optimization Problems

Strategies of solving the problems:

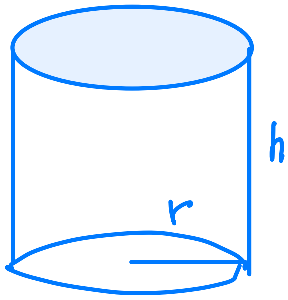
1. Understand the problem.
2. Draw a diagram.
3. Introduce notations.
4. Express the concerning quantity, Q, in terms of other quantities.
5. Eliminate quantities if necessary so that Q is a function, of one variable, say $f(x)$. Determine the domain of this function.
6. Find the absolute extreme values of $f(x)$.

First Derivative Test for Absolute Extreme Values

Suppose that c is a critical number of a continuous function f defined on an interval.
(a) If $f'(x) > 0$ for all $x < c$ and $f'(x) < 0$ for all $x > c$, then $f(c)$ is the absolute maximum value of f.
(b) If $f'(x) < 0$ for all $x < c$ and $f'(x) > 0$ for all $x > c$, then $f(c)$ is the absolute minimum value of f.

Example 1.1

A cylindrical can is to be made to hold $V m^3$ of liquid. Find the dimensions that will minimize the area of the surface of the can.



Let r be the radius of the base of the can. Let h be the height of the can.

$$A = 2\pi r^2 + 2\pi r\cdot h \quad \because \quad V = \pi r^2\cdot h \quad \therefore \quad h = \frac{V}{\pi r^2}.$$

$$\Rightarrow \quad A = 2\pi r^2 + 2\pi r\cdot\frac{V}{\pi r^2} = 2\pi r^2 + \frac{2V}{r} \quad \text{for } r\in(0,\infty)$$

$$\frac{dA}{dr} = 4\pi r - \frac{2V}{r^2} = \frac{2}{r^2}(2\pi r^3 - V). \quad \frac{dA}{dr} = 0 \Rightarrow r = \sqrt[3]{\frac{V}{2\pi}}.$$

$$\frac{dA}{dr} < 0 \text{ for } r\in\left(0, \sqrt[3]{\frac{V}{2\pi}}\right), \quad \frac{dA}{dr} > 0 \text{ for } r\in\left(\sqrt[3]{\frac{V}{2\pi}}, \infty\right)$$

$\Rightarrow$ $A\left(\sqrt[3]{\frac{V}{2\pi}}\right)$ is the abs mini.

Example 1.2

A straight rive is 3 km wide. The point A is on a bank of the river while the point B is on the opposite bank and is 8 km downstream from A. A man can row 6 km/h and run 8 km/h. How can he go from A to B in the shortest time?

Example 1.3

Let $C(x)$ be the cost of producing x units of a commodity. Define the average cost function by $AC(x) = \dfrac{C(x)}{x}$ and the marginal cost function by $MC(x) = \dfrac{d}{dx}C(x)$. Assume that $C(x)$ is differentiable. Prove that the marginal cost function always passes through the minimum point of the average cost function.

Sol: Assume that $AC(x)$ attains the abs min at $x_0 > 0$.

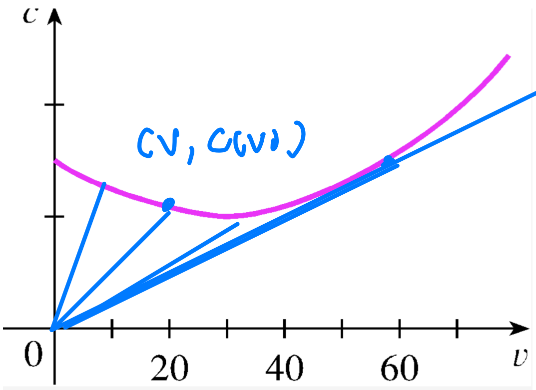
Fermat's Thm $\Rightarrow$ $\left.\dfrac{d}{dx}AC(x)\right|_{x=x_0} = 0$.

$$\frac{d}{dx}AC(x) = \frac{d}{dx}\left(\frac{C(x)}{x}\right) = \frac{C'(x)\cdot x - C(x)}{x^2}$$

$$AC'(x_0) = 0 \Rightarrow C'(x_0)x_0 - C(x_0) = 0 \Rightarrow C'(x_0) = \frac{C(x_0)}{x_0}$$

Example 1.4

The graph shows the fuel consumption of a car (measured in gallons per hour) as a function of the speed v of the car. Using the graph, estimate the speed at which it is the most efficient.



For fixed distance D, the total fuel consumed at speed v is

$$F(v) = c(v)\cdot\frac{D}{v} = D\cdot\frac{c(v)}{v}$$

$$F'(v) = 0 \Rightarrow \underline{c'(v)} = \underline{\frac{c(v)}{v}}.$$

the slope of the tangent line

Example 1.5

Suppose that $R(x)$ denotes the revenue when x units are produced and sold and $C(x)$ represents the cost. Then the profit of selling x units equals to $\Pi(x) = R(x) - C(x)$. Assume that $R(x)$ and $C(x)$ are twice differentiable. Show that if the profit attains a local maximum at $x_0 > 0$, then $R'(x_0) = C'(x_0)$ and $R''(x_0) \leq C''(x_0)$.

Fermat's Thm $\Rightarrow$ $\left.\frac{d}{dx}\Pi(x)\right|_{x=x_0} = 0 = R'(x_0) - C'(x_0)$.

Assume that $R''(x_0) > C''(x_0)$.

Then $\Pi''(x_0) = R''(x_0) - C''(x_0) > 0$. $\Rightarrow$ $\Pi(x_0)$ is a local mini by S.D.T. Contradiction.

Hence $R''(x_0) \leq C''(x_0)$.

Example 1.6

Let $q(p)$ be the demand function where p is the price of the commodity. Then the total revenue is $R(p) = p \cdot q(p)$. Assume that $q(p)$ is differentiable and $R(p)$ attains maximum at $p_0 > 0$. Find the point elasticity at p_0, $\epsilon(p_0) = \dfrac{p_0}{q(p_0)} q'(p_0)$.

Fermat's Thm $\Rightarrow$ $\left.\frac{d}{dp}R\right|_{p=p_0} = 0$

$$\left.\left(q(p) + p \cdot q'(p)\right)\right|_{p=p_0} = 0$$

$\Rightarrow$ $q(p_0) + p_0 q'(p_0) = 0$

$\Rightarrow$ $q(p_0)\left(1 + \dfrac{p_0 \cdot q'(p_0)}{q(p_0)}\right) = 0$ $\Rightarrow$ $\varepsilon(p_0) = -1$.

 $= q(p_0)(1 + \varepsilon(p_0))$

16.2 Antiderivatives

Definition:

A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I.

Theorem:

If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n \ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$\dfrac{1}{x}$	$\ln \lvert x \rvert$	$\dfrac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\dfrac{1}{1+x^2}$	$\tan^{-1} x$
b^x	$\dfrac{b^x}{\ln b}$	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$